miluphcuda

Physical model and numerical methods



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Abstract

This document describes the equations solved by miluphcuda and their discretization: the governing equations of hydrodynamics and continuum mechanics, the equations of state, the strength, fragmentation and porosity models, the SPH discretization including the tensorial correction, the time integrators and the self-gravity. The compile-time switches that select the individual parts are given in `typewriter` font; they are listed in `doc/compile-time-options.md`. How to install and run the code, the command-line options, the file formats, and the material config file are described in the Markdown files in `doc/` and in `material-config/CREATE-MATERIAL-CONFIG.md`.

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1 Introduction

miluphcuda is a smoothed particle hydrodynamics (SPH) code for fluids and for solid and porous bodies, mainly developed for the simulation of impacts and collisions of solid, self-gravitating bodies in planetary science. It is the CUDA port of the code miluph and runs on a single Nvidia GPU. The code and its numerics are described in Schäfer et al. (2016) and Schäfer et al. (2020); please cite both papers and the ASCL record (ascl:1911.023) if you use the code.

The code solves either the Euler equations (`HYDRO`) or the equations of continuum mechanics with deviatoric stress (`SOLID`) in one, two or three dimensions (`DIM`). An arbitrary number of materials with different equations of state and material models can interact. Self-gravity is computed with a Barnes-Hut tree.

Authors. Christoph Schäfer and Sven Riecker wrote the basic version. Later additions, among others: soil model (Samuel Scherrer), Johnson-Cook model (Janka Werner), porosity models (Oliver Wandel), ANEOS support, Collins and Mohr-Coulomb strength models, low-density weakening, rk2 integrator settings (Christoph Burger), Drucker-Prager model, Navier-Stokes equation, tensorial correction, fast integration scheme, return mapping of the p-alpha model, kick-drift-kick integrator (Christoph Schäfer). See `README.md` for the list of developers.

Notation. Particles are labelled i and j, Greek indices denote spatial coordinates and run from 1 to D (the dimension), and the Einstein summation convention applies to them. $\mathbf{x}_{ij} = \mathbf{x}_i - \mathbf{x}_j$, $\mathbf{v}_{ij} = \mathbf{v}_i - \mathbf{v}_j$, $r_{ij} = |\mathbf{x}_{ij}|$, and $\nabla_i W_{ij} = \partial W(r_{ij}, h)/\partial \mathbf{x}_i$. e is the specific internal energy. In miluphcuda the smoothing length h is the radius of the kernel support, i.e. $W(r, h) = 0$ for $r \geq h$.

2 Governing equations

2.1 Conservation equations

The continuity equation, the momentum equation and the energy equation in Lagrangian form are

$$\frac{\mathrm{d}\varrho}{\mathrm{d}t} = -\varrho \frac{\partial v^\alpha}{\partial x^\alpha}, \tag{1}$$

$$\frac{\mathrm{d}v^\alpha}{\mathrm{d}t} = \frac{1}{\varrho}\frac{\partial \sigma^{\alpha\beta}}{\partial x^\beta}, \tag{2}$$

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{1}{\varrho}\sigma^{\alpha\beta}\dot{\varepsilon}^{\alpha\beta} = -\frac{p}{\varrho}\frac{\partial v^\alpha}{\partial x^\alpha} + \frac{1}{\varrho}S^{\alpha\beta}\dot{\varepsilon}^{\alpha\beta}, \tag{3}$$

with the stress tensor

$$\sigma^{\alpha\beta} = -p\,\delta^{\alpha\beta} + S^{\alpha\beta}, \tag{4}$$

the pressure p, the deviatoric stress $S^{\alpha\beta}$ (zero for `HYDRO`), and the strain rate tensor

$$\dot{\varepsilon}^{\alpha\beta} = \frac{1}{2}\left(\frac{\partial v^\alpha}{\partial x^\beta} + \frac{\partial v^\beta}{\partial x^\alpha}\right). \tag{5}$$

The continuity equation is integrated if `INTEGRATE_DENSITY` is set; otherwise the density is computed as the kernel sum $\varrho_i = \sum_j m_j W_{ij}$. With `INTEGRATE_DENSITY`, individual materials can still use the kernel sum (`density_via_kernel_sum` in the material config). The energy equation is integrated if `INTEGRATE_ENERGY` is set.

2.2 Constitutive equations

For an elastic solid, the deviatoric stress evolves according to Hooke's law in rate form, made frame-indifferent with the Jaumann rate,

$$\frac{\mathrm{d}S^{\alpha\beta}}{\mathrm{d}t} = 2\mu\left(\dot{\varepsilon}^{\alpha\beta} - \frac{1}{3}\delta^{\alpha\beta}\dot{\varepsilon}^{\gamma\gamma}\right) + S^{\alpha\gamma}R^{\beta\gamma} + S^{\beta\gamma}R^{\alpha\gamma}, \tag{6}$$

with the shear modulus μ and the rotation rate tensor

$$R^{\alpha\beta} = \frac{1}{2}\left(\frac{\partial v^\alpha}{\partial x^\beta} - \frac{\partial v^\beta}{\partial x^\alpha}\right). \tag{7}$$

The pressure follows from the equation of state (Sect. 3). The deviatoric stress is limited by the plasticity models (Sect. 4) and reduced by damage (Sect. 5).

2.3 Physical viscosity

With `NAVIER_STOKES`, a viscous stress tensor

$$T^{\alpha\beta} = \eta\left(\frac{\partial v^\alpha}{\partial x^\beta} + \frac{\partial v^\beta}{\partial x^\alpha} - \frac{2}{3}\delta^{\alpha\beta}\frac{\partial v^\gamma}{\partial x^\gamma}\right) \tag{8}$$

is added to the momentum equation, with $\eta = \varrho\nu$ and either a constant kinematic viscosity ν (`CONSTANT_KINEMATIC_VISCOSITY`) or ν from the α prescription of Shakura & Sunyaev (`SHAKURA_SUNYAEV_ALPHA`), using the disc scale height from the material config. `KLEY_VISCOSITY` adds the artificial bulk viscosity of Schäfer et al. (2004).

3 Equations of state

The equation of state (EOS) of each material is set by `eos.type` in the material config file. The following types are available (`include/pressure.h`).

type	name	description
−1	`EOS_TYPE_IGNORE`	particle is deactivated
0	`EOS_TYPE_POLYTROPIC_GAS`	$p = K\varrho^\gamma$
1	`EOS_TYPE_MURNAGHAN`	Murnaghan, Sect. 3.1
2	`EOS_TYPE_TILLOTSON`	Tillotson, Sect. 3.2
3	`EOS_TYPE_ISOTHERMAL_GAS`	$p = c_\mathrm{s}^2\varrho$, constant c_s
4	`EOS_TYPE_REGOLITH`	soil model of Bui et al. (2008), experimental
5	`EOS_TYPE_JUTZI`	p-alpha porosity, Tillotson matrix, Sect. 6.1
6	`EOS_TYPE_JUTZI_MURNAGHAN`	p-alpha porosity, Murnaghan matrix
7	`EOS_TYPE_ANEOS`	tabulated EOS (ANEOS), Sect. 3.3
8	`EOS_TYPE_VISCOUS_REGOLITH`	experimental, do not use
9	`EOS_TYPE_IDEAL_GAS`	$p = (\gamma - 1)\varrho e$
10	`EOS_TYPE_SIRONO`	Sirono porosity model
11	`EOS_TYPE_EPSILON`	ε-α porosity, Tillotson matrix
12	`EOS_TYPE_LOCALLY_ISOTHERMAL_GAS`	$p = c_\mathrm{s}^2\varrho$, $c_\mathrm{s} = v_\mathrm{K}(r)\,H/r$ for discs around point masses
13	`EOS_TYPE_JUTZI_ANEOS`	p-alpha porosity, ANEOS matrix

3.1 Murnaghan

$$p = \frac{K_0}{n}\left[\left(\frac{\varrho}{\varrho_0}\right)^n - 1\right], \qquad c_\mathrm{s}^2 = \frac{K_0}{\varrho_0}\left(\frac{\varrho}{\varrho_0}\right)^{n-1}, \tag{9}$$

with the bulk modulus K_0 and the reference density ϱ_0 (Melosh, 1996). For $n = 1$ this is the liquid EOS $p = c_\mathrm{s}^2(\varrho - \varrho_0)$. For $\varrho/\varrho_0 < \eta_\mathrm{limit}$ (`rho_limit`, default 0.9) the pressure is set to zero, since the material would fragment or form droplets instead of sustaining large tension. The sound speed is limited from below by `cs_limit`.

3.2 Tillotson

The Tillotson EOS (Tillotson, 1962; Melosh, 1996) uses $\eta = \varrho/\varrho_0$, $\chi = \eta - 1$, $z = \varrho_0/\varrho - 1$ and $x = e/(E_0\eta^2)$. In the compressed state the pressure is

$$p_\mathrm{c} = \left(a + \frac{b}{x+1}\right)\varrho e + A\chi + B\chi^2, \tag{10}$$

in the expanded state

$$p_\mathrm{e} = a\varrho e + \left(\frac{b\varrho e}{x+1} + A\chi\,\mathrm{e}^{-\beta z}\right)\mathrm{e}^{-\alpha z^2}. \tag{11}$$

The regimes are

- $\eta < \eta_\mathrm{limit}$ and $e \le E_\mathrm{iv}$ (cold, expanded, fragmented): $p = 0$;
- $\eta \ge 1$ or $e \le E_\mathrm{iv}$: $p = p_\mathrm{c}$;
- $e \ge E_\mathrm{cv}$ (and $\eta < 1$): $p = p_\mathrm{e}$;
- $E_\mathrm{iv} < e < E_\mathrm{cv}$ (and $\eta < 1$): linear interpolation in e, $p = [(E_\mathrm{cv} - e)\,p_\mathrm{c} + (e - E_\mathrm{iv})\,p_\mathrm{e}]/(E_\mathrm{cv} - E_\mathrm{iv})$, where p_c is set to zero for $\eta < \eta_\mathrm{limit}$, so that p is continuous in e.

The sound speed is computed consistently from the analytic derivatives, $c_\mathrm{s}^2 = \partial p/\partial\varrho|_e + (p/\varrho^2)\,\partial p/\partial e|_\varrho$, and limited from below by `cs_limit` (default 1 % of $\sqrt{A/\varrho_0}$). For type 2, e is clamped to $e \ge 0$ in the EOS. If `polytropic_gamma` > 1 is set, completely vaporized material ($e > 100E_\mathrm{cv}$ and $\eta < 1$) is treated as an ideal gas, $p = (\gamma - 1)\varrho e$.

3.3 ANEOS and tabulated EOS

For type 7 (and as matrix EOS for type 13) pressure and sound speed are interpolated bilinearly in a table $p(\varrho, e)$, $c_\mathrm{s}(\varrho, e)$, which is typically generated with ANEOS (e.g. with the external project miluphA-NEOS). Energies below the table are clamped to the lowest table energy (cold curve). Above the table maximum the ideal-gas fallback $p = (\gamma_\mathrm{A} - 1)\varrho e$ with `aneos_gamma` is used. With `MORE_ANEOS_OUTPUT` or `ANEOS_VAPOR_NO_STRENGTH`, the tables of temperature, entropy and the ANEOS phase flag are read as well.

Vapour without strength. With `ANEOS_VAPOR_NO_STRENGTH`, particles of ANEOS materials whose phase flag is `ANEOS_PHASE_TWO_PHASE_LV` (two-phase liquid/vapour, also assigned in the ideal-gas fallback) get $S^{\alpha\beta} = 0$ and $p \ge 0$ when the stress tensor is assembled (`src/stress.cu`). Since the stored deviatoric stress is set to zero as well, a particle that leaves the two-phase region starts with zero deviatoric stress.

3.4 Gases

The ideal gas (type 9), the polytropic gas (type 0), and the isothermal gases (types 3, 12) are listed in the table above; the sound speed of the ideal gas is $c_\mathrm{s} = \sqrt{\gamma p/\varrho}$. With `REAL_HYDRO`, negative pressures are set to zero for all materials.

4 Strength models

The plasticity models limit the deviatoric stress after each evaluation of the right-hand side. With the second invariant $J_2 = \frac{1}{2} S^{\alpha\beta} S^{\alpha\beta}$ and the yield strength Y, the deviatoric stress is scaled by

$$S^{\alpha\beta} \to f\, S^{\alpha\beta}, \qquad f = \min\left(\frac{Y}{\sqrt{J_2}}, 1\right) \tag{12}$$

for the pressure-dependent models (Drucker-Prager, Mohr-Coulomb, Collins), and by

$$f = \min\left(\frac{Y^2}{3J_2}, 1\right) \tag{13}$$

for the von Mises model (von Mises, 1913; Benz and Asphaug, 1995). The pressure-dependent models compare Y with $\sqrt{J_2}$, i.e. Y is the yield strength in shear.

4.1 Von Mises

$Y = Y_0$ (`yield_stress`) is constant. `VON_MISES_PLASTICITY` can additionally be combined with Drucker-Prager or Mohr-Coulomb, in which case Y_0 is an upper limit, $Y \to \min(Y, Y_0)$.

4.2 Drucker-Prager

$$Y = A + 3Bp, \qquad A = \frac{6\,c\cos\phi}{\sqrt{3}\,(3-\sin\phi)}, \qquad B = \frac{2\sin\phi}{\sqrt{3}\,(3-\sin\phi)}, \tag{14}$$

with the cohesion c and the angle of internal friction ϕ, and $Y \geq 0$. Negative pressures are not capped.

4.3 Mohr-Coulomb

With the coefficient of internal friction $\mu_\mathrm{i} = \tan\phi$,

$$Y = \begin{cases} c + \mu_\mathrm{i} p & p > 0, \\ c + p & -c < p \leq 0, \\ 0 & p \leq -c, \end{cases} \tag{15}$$

and negative pressures are capped at $p \geq -c$.

4.4 Collins

Following Collins et al. (2004) and the implementation of Jutzi (2015), the yield strength of intact material follows the Lundborg curve (Lundborg, 1968),

$$Y_\mathrm{i} = c + \frac{\mu_\mathrm{i} p}{1 + \mu_\mathrm{i} p/(Y_\mathrm{M} - c)} \quad (p > 0), \qquad Y_\mathrm{i} = c \quad (p \leq 0), \tag{16}$$

with the limit Y_M (`yield_stress`) for $p \to \infty$. With `COLLINS_PLASTICITY_INCLUDE_MELT_ENERGY`, Y_i is multiplied by $(1-e/e_\mathrm{melt})$ (and zero for $e \geq e_\mathrm{melt}$). With `FRAGMENTATION`, the strength of damaged material is

$$Y_\mathrm{d} = \begin{cases} c_\mathrm{d} + \mu_\mathrm{d} p & p > 0, \\ c_\mathrm{d} + p & -c_\mathrm{d} < p \leq 0, \\ 0 & p \leq -c_\mathrm{d}, \end{cases} \tag{17}$$

and the yield strength is $Y = \min[(1-D)Y_\mathrm{i} + D\,Y_\mathrm{d}, Y_\mathrm{i}]$ with the damage D. Negative pressure is released by damage, $p \to (1-D)p$, but only down to the residual tensile strength $p_\mathrm{t} = \max(c_\mathrm{d}, p_\mathrm{tl})$ of fully damaged material (p_tl: `tensile_pressure_limit`): for $p < -p_\mathrm{t}$, p is replaced by $\min[(1-D)p, -p_\mathrm{t}]$; pressures $p \geq -p_\mathrm{t}$ are not changed.

4.5 Collins simple

`COLLINS_PLASTICITY_SIMPLE` uses only the Lundborg curve for $p > 0$, $Y = c + p$ for $-c < p \leq 0$, $Y = 0$ below, and caps negative pressures at $p \geq -c$. It is not combined with the damage model.

4.6 Low-density weakening

With `LOW_DENSITY_WEAKENING`, the cohesion (for von Mises: the yield strength; for Collins: the damaged cohesion) is multiplied by

$$f_\mathrm{ldw} = \begin{cases} 1 & \eta \geq 1, \\ \left(\dfrac{\eta - \eta_\mathrm{l}}{1 - \eta_\mathrm{l}}\right)^a (1-\gamma) + \gamma & \eta_\mathrm{l} < \eta < 1, \\ \left(\dfrac{\eta}{\eta_\mathrm{l}}\right)^b \gamma & \eta \leq \eta_\mathrm{l}, \end{cases} \tag{18}$$

with $\eta = \varrho/\varrho_0$ (matrix density for porous materials) and the parameters η_l, a, b, γ of the group `low_density_weakening`.

4.7 Plastic strain and total strain

For the standard models, the equivalent plastic strain is accumulated from the radial return,

$$\Delta\varepsilon_\mathrm{p} = (1-f)\,\frac{\sqrt{3J_2}}{3G_\mathrm{eff}}, \tag{19}$$

with J_2 of the trial stress and the shear modulus G_eff with which S grows in eq. (6) (μ/α for p-alpha materials). Independently of the plasticity model, the total integrated strain

$$\frac{\mathrm{d}\varepsilon_\mathrm{tot}}{\mathrm{d}t} = \sqrt{\frac{2}{3}\,\dot{\varepsilon}'^{\alpha\beta}\dot{\varepsilon}'^{\alpha\beta}}, \qquad \dot{\varepsilon}'^{\alpha\beta} = \dot{\varepsilon}^{\alpha\beta} - \frac{1}{3}\delta^{\alpha\beta}\dot{\varepsilon}^{\gamma\gamma}, \tag{20}$$

is integrated for `SOLID` (HDF5 output `/total_integrated_strain`).

4.8 Experimental models

`JC_PLASTICITY` (Johnson-Cook), `PURE_REGOLITH` (Bui et al., 2008) and `VISCOUS_REGOLITH` are experimental and not described here.

5 Fragmentation

With FRAGMENTATION, brittle failure under tension is modelled with the Grady-Kipp model (Grady and Kipp, 1980) as implemented by Benz and Asphaug (1995). The scalar damage $0 \leq D \leq 1$ parametrizes the influence of cracks: $D = 0$ is intact, $D = 1$ fully damaged material that cannot sustain tension. Damage reduces the stress,

$$\sigma^{\alpha\beta} = -\hat{p}\,\delta^{\alpha\beta} + (1-D)\,S^{\alpha\beta}, \qquad \hat{p} = \begin{cases} p & p \geq 0, \\ (1-D)\,p & p < 0, \end{cases} \tag{21}$$

where the reduction of $S^{\alpha\beta}$ is applied only with DAMAGE_ACTS_ON_S and the reduction of negative pressure is done in the plasticity model for COLLINS_PLASTICITY. The stored $S^{\alpha\beta}$ is the undamaged deviator; damage is applied where the stress is assembled.

The number of flaws per unit volume with activation strains below ε follows the Weibull distribution $n(\varepsilon) = k\varepsilon^m$ (Weibull, 1939). The flaws are distributed among the particles in a preprocessing step (Benz and Asphaug, 1995): for flaw $j = 1, \ldots, N_{\text{flaw}}$ a random particle gets the activation threshold $\varepsilon_j = [j/(kV)]^{1/m}$ until each particle has at least one flaw, with the volume V of the brittle body; on average $N_{\text{flaw}} = N \ln N$. The thresholds are part of the input file.

The local scalar strain is computed from the largest principal stress $\sigma_{\max}$ of the (damage-reduced) stress tensor,

$$\varepsilon = \frac{\sigma_{\max}}{(1-D)\,E}, \qquad E = \frac{9K\mu}{3K+\mu}. \tag{22}$$

All flaws with thresholds below ε are active, and the damage grows as

$$\frac{\mathrm{d}D^{1/D_{\text{dim}}}}{\mathrm{d}t} = n_{\text{act}}\,\frac{c_{\text{g}}}{h}, \qquad c_{\text{g}} = 0.4\sqrt{\frac{K + 4/3(1-D)\,\mu}{\varrho}}, \tag{23}$$

where n_{act} is the number of active flaws, D_{dim} the dimension, and c_{g} the crack growth speed (40 % of the longitudinal wave speed in the damaged material). The integrated variable is $D^{1/D_{\text{dim}}}$. It is limited by $(n_{\text{act}}/n_{\text{tot}})^{1/D_{\text{dim}}}$, so that a particle can only be fully damaged if all its flaws are active. For porous materials, the porous damage D_{p} (Sect. 6.1) is added: $D = \min(1, D_{\text{t}} + D_{\text{p}})$ with the tensile damage D_{t}.

6 Porosity

6.1 The p-alpha model

The p-alpha model (Herrmann, 1969) as formulated for SPH by Jutzi et al. (2008) describes a porous material by the distention $\alpha = \varrho_{\text{s}}/\varrho \geq 1$, with the matrix density ϱ_{s} and the bulk density ϱ. The pressure is

$$P = \frac{1}{\alpha}\,P_{\text{s}}(\alpha\varrho, e), \tag{24}$$

where P_{s} is the matrix EOS: Tillotson (type 5), Murnaghan (type 6) or ANEOS (type 13), evaluated at the matrix density.

Crush curves. Compaction follows the crush curve $\alpha_{\mathrm{cc}}(P)$ with $\alpha_{\mathrm{cc}} = \alpha_0$ for $P \le p_{\mathrm{e}}$ and $\alpha_{\mathrm{cc}} = 1$ for $P \ge p_{\mathrm{s}}$. In between,

$$\text{style 0:}\quad \alpha_{\mathrm{cc}}(P) = 1 + (\alpha_0 - 1)\frac{(p_{\mathrm{s}} - P)^2}{(p_{\mathrm{s}} - p_{\mathrm{e}})^2}, \tag{25}$$

$$\text{style 1:}\quad \alpha_{\mathrm{cc}}(P) = 1 + s(\alpha_{\mathrm{t}} - 1)\left(\frac{p_{\mathrm{s}} - P}{p_{\mathrm{s}} - p_{\mathrm{e}}}\right)^{n_2} + s(\alpha_{\mathrm{e}} - \alpha_{\mathrm{t}})\left(\frac{p_{\mathrm{t}} - P}{p_{\mathrm{t}} - p_{\mathrm{e}}}\right)^{n_1}\Theta(p_{\mathrm{t}} - P), \tag{26}$$

with $s = (\alpha_0 - 1)/(\alpha_{\mathrm{e}} - 1)$, which rescales the curve of Jutzi et al. (2008) such that $\alpha_{\mathrm{cc}}(p_{\mathrm{e}}) = \alpha_0$. Styles 2 and 4 are experimental crush curves with hard-coded constants (`src/pressure.cu`).

Time evolution of α. On the crush curve, $\dot\alpha = (\mathrm{d}\alpha/\mathrm{d}P)\,\dot P$ with $P = P_{\mathrm{s}}(\alpha\varrho, e)/\alpha$, which gives (Jutzi et al., 2008, eq. 25)

$$\frac{\mathrm{d}\alpha}{\mathrm{d}t} = \frac{\left(\dot e\,\dfrac{\partial P_{\mathrm{s}}}{\partial e} + \alpha\,\dot\varrho\,\dfrac{\partial P_{\mathrm{s}}}{\partial \varrho_{\mathrm{s}}}\right)\dfrac{\mathrm{d}\alpha}{\mathrm{d}P}}{\alpha + \dfrac{\mathrm{d}\alpha}{\mathrm{d}P}\left(P - \varrho\,\dfrac{\partial P_{\mathrm{s}}}{\partial \varrho_{\mathrm{s}}}\right)}. \tag{27}$$

Unloading is elastic: if eq. (27) gives $\dot\alpha > 0$ (pressure decreasing), $\dot\alpha = 0$. With `PALPHA_ELASTIC_RELOADING`, reloading below the crush curve is elastic as well: for $\alpha < \alpha_{\mathrm{cc}}(P) - \epsilon_{\mathrm{tol}}$, $\mathrm{d}\alpha/\mathrm{d}P = 0$. Then $\alpha = \alpha_{\mathrm{cc}}(P_{\max})$ keeps the memory of the maximum pressure. Without it, every pressure increase above p_{e} compacts (Jutzi et al., 2008, eqs. 22–23).

Return mapping. With `PALPHA_RETURN_MAPPING`, α is not integrated. For each evaluated state, α is determined from ϱ and e with the distention α_{h} after the last accepted step: if $g(\alpha_{\mathrm{h}}) > 0$ with

$$g(\alpha) = \alpha - \alpha_{\mathrm{cc}}(P_{\mathrm{s}}(\alpha\varrho, e)/\alpha)\,, \tag{28}$$

the state is compressed beyond the crush curve and α is the root of g in $[1, \alpha_{\mathrm{h}}]$ (Newton iteration safeguarded by bisection); otherwise $\alpha = \alpha_{\mathrm{h}}$. Since $\alpha\,\mathrm{d}g/\mathrm{d}\alpha$ is the denominator of eq. (27), g is increasing and the root unique wherever the time integration of α is well defined. α then follows the crush curve independently of the time step. The changes of the deviatoric stress and of the porous damage that the integration of $\dot\alpha$ would cause are applied in closed form.

Deviatoric stress. With `STRESS_PALPHA_POROSITY`, the code integrates the porous deviator $\hat S^{\alpha\beta} = S^{\alpha\beta}/\alpha$ with the matrix deviator $S^{\alpha\beta}$ (Jutzi et al., 2008, eqs. 45, 49),

$$\frac{\mathrm{d}\hat S^{\alpha\beta}}{\mathrm{d}t} = \frac{f_{\mathrm{r}}}{\alpha}\,2\mu\left(\dot\varepsilon^{\alpha\beta} - \frac{1}{3}\delta^{\alpha\beta}\dot\varepsilon^{\gamma\gamma}\right) + f_{\mathrm{r}}\left(\hat S^{\alpha\gamma}R^{\beta\gamma} + \hat S^{\beta\gamma}R^{\alpha\gamma}\right) - \frac{\hat S^{\alpha\beta}}{\alpha}\frac{\mathrm{d}\alpha}{\mathrm{d}t}, \tag{29}$$

with $f_{\mathrm{r}} = f$ for `PALPHA_F_ACTS_ON_ROTATION` = 1 (as in Jutzi et al. 2008) and $f_{\mathrm{r}} = 1$ otherwise. The factor

$$f = 1 + \frac{\varrho}{\alpha}\frac{\mathrm{d}\alpha}{\mathrm{d}\varrho}, \qquad \frac{\mathrm{d}\alpha}{\mathrm{d}\varrho} = \frac{\left(\dfrac{P}{\varrho^2}\dfrac{\partial P_{\mathrm{s}}}{\partial e} + \alpha\dfrac{\partial P_{\mathrm{s}}}{\partial \varrho_{\mathrm{s}}}\right)\dfrac{\mathrm{d}\alpha}{\mathrm{d}P}}{\alpha + \dfrac{\mathrm{d}\alpha}{\mathrm{d}P}\left(P - \varrho\,\dfrac{\partial P_{\mathrm{s}}}{\partial \varrho_{\mathrm{s}}}\right)}, \tag{30}$$

relates the volumetric strain rate of the matrix to that of the bulk material; $f \le 1$, and $f = 1$ for elastic states. Physically, f belongs only to the volumetric part of the strain rate, which drops out of the traceless Hooke term; therefore `PALPHA_F_ACTS_ON_ROTATION` = 0 ($f_{\mathrm{r}} = 1$) is recommended. It leaves the Jaumann terms, which are pure kinematics, unscaled and reproduces the rigid rotation of a porous particle exactly.

Porous damage. With `FRAGMENTATION`, compaction produces damage (Jutzi et al., 2008, eq. 57),

$$D_{\mathrm{p}}^{1/D_{\mathrm{dim}}} = \frac{\left(1 - \dfrac{\alpha - 1}{\alpha_0 - 1} + \delta\right)^{1/D_{\mathrm{dim}}} - \delta^{1/D_{\mathrm{dim}}}}{(1+\delta)^{1/D_{\mathrm{dim}}} - \delta^{1/D_{\mathrm{dim}}}}, \qquad \delta = 0.01, \tag{31}$$

which is integrated in time along with α (or applied exactly with the return mapping).

Sound speed. The sound speed is interpolated linearly in α between the sound speed of the matrix ($\alpha = 1$) and c_{porous} (`cs_porous`, $\alpha = \alpha_0$).

6.2 The ε-α model

With `EPSALPHA_POROSITY` (type 11), the distention is a function of the volumetric strain ε_V (Wünnemann et al., 2006): elastic for $\varepsilon_V \geq \varepsilon_{\mathrm{e}}$, exponential compaction $\mathrm{d}\alpha/\mathrm{d}\varepsilon_V = \alpha_0 \kappa\, \mathrm{e}^{\kappa(\varepsilon_V - \varepsilon_{\mathrm{e}})}$ for $\varepsilon_{\mathrm{x}} \leq \varepsilon_V < \varepsilon_{\mathrm{e}}$, a power law down to full compaction at ε_{c}. The pressure is $P = P_{\mathrm{s}}(\alpha\varrho, e)/\alpha$ with the Tillotson EOS.

6.3 The Sirono model

With `SIRONO_POROSITY` (type 10), the modified Sirono model of Geretshauser et al. (2011) for highly porous dust aggregates is used, with density-dependent compressive, tensile and shear strength.

7 SPH discretization

7.1 Kernels

The kernel is chosen at run time with `-k`. With $q = r/h$ and $(\cdot)_+ = \max(0, \cdot)$:

Cubic spline (Monaghan and Lattanzio, 1985), `cubic_spline` (default),

$$W = \frac{\sigma}{h^D} \begin{cases} 6q^3 - 6q^2 + 1 & 0 \leq q < \frac{1}{2}, \\ 2(1-q)^3 & \frac{1}{2} \leq q \leq 1, \end{cases} \qquad \sigma = \tfrac{4}{3},\ \tfrac{40}{7\pi},\ \tfrac{8}{\pi}. \tag{32}$$

Quintic spline , `quintic_spline`,

$$W = \frac{\sigma}{h^D}\left[(3-3q)_+^5 - 6(2-3q)_+^5 + 15(1-3q)_+^5\right], \qquad \sigma = \tfrac{1}{40},\ \tfrac{63}{478\pi},\ \tfrac{9}{40\pi}. \tag{33}$$

Wendland kernels (Dehnen and Aly, 2012), `wendlandc2`, `wendlandc4`, `wendlandc6`, in 2D and 3D

$$W_{\mathrm{C2}} = \frac{\sigma}{h^D}(1-q)_+^4(1+4q), \qquad \sigma = \tfrac{7}{\pi},\ \tfrac{21}{2\pi}, \tag{34}$$

$$W_{\mathrm{C4}} = \frac{\sigma}{h^D}(1-q)_+^6\left(1 + 6q + \tfrac{35}{3}q^2\right), \qquad \sigma = \tfrac{9}{\pi},\ \tfrac{495}{32\pi}, \tag{35}$$

$$W_{\mathrm{C6}} = \frac{\sigma}{h^D}(1-q)_+^8(1 + 8q + 25q^2 + 32q^3), \qquad \sigma = \tfrac{78}{7\pi},\ \tfrac{1365}{64\pi}, \tag{36}$$

and in 1D $(1-q)_+^3(1+3q)$, $(1-q)_+^5(1+5q+8q^2)$, $(1-q)_+^7(1+7q+19q^2+21q^3)$ with $\sigma = \tfrac{5}{4}, \tfrac{3}{2}, \tfrac{55}{32}$.

Spiky kernel (Desbrun and Cani, 1996), `spiky`, $W = \sigma h^{-D}(1-q)_+^3$ with $\sigma = \frac{10}{\pi}, \frac{15}{\pi}$ in 2D and 3D. The normalization constants are given for $D = 1, 2, 3$ unless noted otherwise.

7.2 Smoothing length and neighbours

For an interaction of particles i and j, the kernel is evaluated with $h_{ij} = (h_i + h_j)/2$, or as $W_{ij} = [W(r_{ij}, h_i) + W(r_{ij}, h_j)]/2$ with `AVERAGE_KERNELS`. The smoothing length is constant unless `VARIABLE_SML` is set, in which case either the number of interaction partners is kept fixed (`FIXED_NOI`, Hernquist and Katz 1989) or

$$\frac{\mathrm{d}h_i}{\mathrm{d}t} = \frac{h_i}{D}\nabla\cdot\mathbf{v}_i = -\frac{h_i}{D}\sum_j \frac{m_j}{\varrho_j}\,\mathbf{v}_{ij}\cdot\nabla_i W_{ij} \tag{37}$$

is integrated (`INTEGRATE_SML`), with h limited to $[f_{\min}h_0, f_{\max}h_0]$ (`factor_sml_min`, `factor_sml_max`). The interaction partners are found with the Barnes-Hut tree; their number is limited by `MAX_NUM_INTERACTIONS`.

7.3 Continuity equation

$$\frac{\mathrm{d}\varrho_i}{\mathrm{d}t} = \sum_j m_j \frac{\varrho_i}{\varrho_j}\,\mathbf{v}_{ij}\cdot\nabla_i W_{ij} \tag{38}$$

(Randles and Libersky, 1996). With `TENSORIAL_CORRECTION_FOR_DRHODT`, $\nabla_i W_{ij}$ is replaced by the corrected gradient $\mathsf{B}_i\nabla_i W_{ij}$ (Sect. 7.9), and ϱ_j by $\tilde{\varrho}_{ij}$, eq. (46).

7.4 Momentum equation

Two SPH representations are available (`SPH_EQU_VERSION`),

$$1:\quad \frac{\mathrm{d}v_i^\alpha}{\mathrm{d}t} = \sum_j m_j\left(\frac{\sigma_i^{\alpha\beta}}{\varrho_i^2}G_i^\beta + \frac{\sigma_j^{\alpha\beta}}{\varrho_j^2}G_j^\beta\right) - \sum_j m_j\Pi_{ij}\,\partial_{i,\alpha}W_{ij}, \tag{39}$$

$$2:\quad \frac{\mathrm{d}v_i^\alpha}{\mathrm{d}t} = \sum_j m_j\frac{\sigma_i^{\alpha\beta}+\sigma_j^{\alpha\beta}}{\varrho_i\varrho_j}\frac{G_i^\beta + G_j^\beta}{2} - \sum_j m_j\Pi_{ij}\,\partial_{i,\alpha}W_{ij}, \tag{40}$$

where $\mathbf{G}_i = \mathsf{B}_i\nabla_i W_{ij}$ and $\mathbf{G}_j = \mathsf{B}_j\nabla_i W_{ij}$ are the corrected kernel gradients with the correction matrices of particles i and j (Sect. 7.9), or $\mathbf{G}_i = \mathbf{G}_j = \nabla_i W_{ij}$ without `TENSORIAL_CORRECTION`, and Π_{ij} is the artificial viscosity. For `HYDRO`, $\sigma^{\alpha\beta} = -p\,\delta^{\alpha\beta}$. The stress tensor includes the reduction by damage. With `ARTIFICIAL_STRESS`, the term $m_j f_{ij}^n(R_i^{\alpha\beta}/\varrho_i^2 G_i^\beta + R_j^{\alpha\beta}/\varrho_j^2 G_j^\beta)$ with $f_{ij} = W(r_{ij})/W(\Delta p)$ is added (Monaghan, 2000; Gray et al., 2001).

7.5 Energy equation

The energy equation is discretized pairwise, consistent with the momentum equation: with the acceleration $\mathbf{a}_{ij}$ of particle i due to particle j (the summand of the momentum equation without the artificial viscosity),

$$\frac{\mathrm{d}e_i}{\mathrm{d}t} = -\frac{1}{2}\sum_j \mathbf{v}_{ij}\cdot\mathbf{a}_{ij} + \frac{1}{2}\sum_j m_j\Pi_{ij}\,\mathbf{v}_{ij}\cdot\nabla_i W_{ij}. \tag{41}$$

For `SPH_EQU_VERSION` 1 and `HYDRO` this is the standard $\dot{e}_i = \frac{1}{2}\sum_j m_j(p_i/\varrho_i^2 + p_j/\varrho_j^2)\,\mathbf{v}_{ij}\cdot\nabla_i W_{ij} + \ldots$. Together with the momentum equation the total energy is conserved.

7.6 Velocity gradient

The strain rate and rotation rate tensors are computed from the velocity gradient. Without tensorial correction (Benz and Asphaug, 1995),

$$\frac{\partial v_i^\alpha}{\partial x^\beta} = -\frac{1}{\varrho_i} \sum_j m_j \, v_{ij}^\alpha \, \partial_{i,\beta} W_{ij}, \tag{42}$$

and with tensorial correction

$$\frac{\partial v_i^\alpha}{\partial x^\beta} = -\sum_j \frac{m_j}{\tilde{\varrho}_{ij}} \, v_{ij}^\alpha \, \left(\mathsf{B}_i \nabla_i W_{ij}\right)^\beta , \tag{43}$$

which is exact for linear velocity fields. $\dot{\varepsilon}$ and R are the symmetric and antisymmetric parts.

7.7 Artificial viscosity

With `ARTIFICIAL_VISCOSITY` (Monaghan and Lattanzio, 1985),

$$\Pi_{ij} = \begin{cases} \dfrac{-\alpha \, \bar{c}_{ij} \mu_{ij} + \beta \mu_{ij}^2}{\bar{\varrho}_{ij}} & \mathbf{v}_{ij} \cdot \mathbf{x}_{ij} < 0, \\ 0 & \text{otherwise}, \end{cases} \qquad \mu_{ij} = \frac{\bar{h}_{ij} \, \mathbf{v}_{ij} \cdot \mathbf{x}_{ij}}{r_{ij}^2 + \epsilon \bar{h}_{ij}^2}, \tag{44}$$

with the means $\bar{c}_{ij}$, $\bar{\varrho}_{ij}$, $\bar{h}_{ij}$ of the sound speeds, densities and smoothing lengths, $\epsilon = 0.01$, and the material parameters α, β (default 1 and 2). With `BALSARA_SWITCH` (Balsara, 1995), μ_{ij} is multiplied by $(f_i + f_j)/2$, $f_i = |\nabla \cdot \mathbf{v}_i| / (|\nabla \cdot \mathbf{v}_i| + |\nabla \times \mathbf{v}_i| + 10^{-4} c_i / h_i)$. `INVISCID_SPH` uses time-dependent coefficients (Cullen and Dehnen, 2010).

7.8 XSPH and Shepard correction

With `XSPH` the particles move with the smoothed velocity $\dot{\mathbf{x}}_i = \mathbf{v}_i + \frac{1}{2} \sum_j m_j / \bar{\varrho}_{ij} \, (\mathbf{v}_j - \mathbf{v}_i) W_{ij}$ (Monaghan, 1989). `SHEPARD_CORRECTION` normalizes the kernel to zeroth-order consistency.

7.9 Tensorial correction

Standard SPH kernel gradients do not reproduce the gradient of a linear field, and solid bodies cannot rotate rigidly without spurious stress. With `TENSORIAL_CORRECTION` (Randles and Libersky, 1996), each particle gets the correction matrix $\mathsf{B}_i = \mathsf{M}_i^{-1}$ with the (symmetric) moment matrix

$$M_i^{\alpha\beta} = \sum_j \frac{m_j}{\tilde{\varrho}_{ij}} \, x_{ji}^\alpha \, \partial_{i,\beta} W_{ij}, \qquad x_{ji}^\alpha = x_j^\alpha - x_i^\alpha, \tag{45}$$

computed with the same kernel and smoothing length as the force loops, and the weight

$$\tilde{\varrho}_{ij} = \begin{cases} \varrho_j & \text{default (Randles and Libersky, 1996)}, \\ \varrho_i & \texttt{TENSORIAL_CORRECTION_CONSERVE_ANGULAR_MOMENTUM}. \end{cases} \tag{46}$$

The corrected gradient $\mathsf{B}_i \nabla_i W_{ij}$ makes the velocity gradient exact for linear fields. Close to free surfaces and contact zones the moment matrix becomes ill-conditioned. With the default ("old-school") scheme in `src/kernel.cu`, the identity is used instead of B_i if $|\det \mathsf{B}_i| < 0.2$ or > 5, or if an entry of B_i exceeds `MAX_ABS_TENSORIAL_CORRECTION_ENTRY` (5). With `TENSORIAL_CORRECTION_CONSERVE_ANGULAR_MOMENTUM`, the conditioning is judged on the matrix with weight ϱ_j, so that a pure density contrast (e.g. at material interfaces) does not reset the correction.

Angular momentum. With weight ϱ_i and `SPH_EQU_VERSION` 1, the pair forces of the momentum equation are torque-free also with deviatoric stress, variable density and variable smoothing length (wherever B is not reset to the identity), so the total angular momentum is conserved exactly by the spatial discretization; with the `kdk` integrator also in time (up to round-off).

Continuity equation. The correction is used in the momentum and energy equations and in the velocity gradient for $\dot{\varepsilon}$ and R, but by default not in the continuity equation and in $\dot{h}$ (`TENSORIAL_CORRECTION_FOR_DRHODT` = 0): with the corrected divergence the density can decouple from the particle geometry, which leads to a collapse of the time step.

8 Time integration

8.1 Integrators

The integrator is chosen with `-I`. All integrators advance positions, velocities, density, energy, smoothing length, deviatoric stress, damage, distention, and plastic and total strain.

`rk2_adaptive` (default). Each step uses three right-hand side evaluations $k_1 = f(y_n)$, $k_2 = f(y_n + \frac{\Delta t}{2} k_1)$, $k_3 = f(y_n - \Delta t\, k_1 + 2\Delta t\, k_2)$ and advances with the third-order combination $y_{n+1} = y_n + \frac{\Delta t}{6}(k_1 + 4k_2 + k_3)$. The difference to the second-order midpoint solution $y_n + \Delta t\, k_2$ is the error estimate. The maximum relative error of the positions (relative to $\max(|\Delta \mathbf{x}|, 0.1\, h)$), velocities, densities and energies, divided by the accuracy ϵ (`-Q`), decides whether the step is accepted. The new step is $\Delta t_{\text{new}} = 0.9\, \Delta t\, E^{-0.3}$ after an accepted and $\max(0.1, 0.9 E^{-0.25}) \Delta t$ after a rejected step, with E the normalized error, and is additionally limited by the Courant condition and by the damage change (and optionally by the accelerations, the pressure change and the change of α), see `include/rk2adaptive.h`.

`kdk` (kick-drift-kick, leapfrog). With one right-hand side evaluation per step,

$$\mathbf{v}^{n+1/2} = \mathbf{v}^n + \frac{\Delta t}{2} \mathbf{a}^n, \qquad \mathbf{x}^{n+1} = \mathbf{x}^n + \Delta t\, \mathbf{v}^{n+1/2}, \tag{47}$$

$$\mathbf{a}^{n+1} = \mathbf{a}\left(\mathbf{x}^{n+1}, \mathbf{v}^n + \Delta t\, \mathbf{a}^n, Q^n + \Delta t\, \dot{Q}^n\right), \qquad \mathbf{v}^{n+1} = \mathbf{v}^{n+1/2} + \frac{\Delta t}{2} \mathbf{a}^{n+1}, \tag{48}$$

and the same kicks for all other integrated quantities Q. The right-hand side at the end of the step is evaluated at the predicted state, as in Phantom (Price et al., 2018). Since the drift uses the velocity of the kicks, the total angular momentum is conserved to round-off for every step size if the forces are torque-free (`TENSORIAL_CORRECTION_CONSERVE_ANGULAR_MOMENTUM`, no XSPH, no boundaries, no tree gravity). The time step is set before each step from the Courant condition and optional limits (`include/kdk.h`); there is no error control.

Fixed-order schemes. `euler` (first order, fixed Δt given with `-M`), `euler_pc` and `monaghan_pc` (predictor-corrector schemes of second order with predictor step Δt and $\Delta t/2$), and `heun_rk4` (Heun scheme for the SPH particles coupled with a fourth-order Runge-Kutta scheme for gravitating point masses). The predictor-corrector and Heun schemes limit the step by the Courant condition and the accelerations.

8.2 Courant condition

$$\Delta t \le C \min_i \frac{h_i}{\sqrt{c_{\mathrm{s},i}^2 + \frac{4}{3}\mu/\varrho_i}}, \tag{49}$$

with C = `COURANT_FACT` = 0.4 (note that h is the full support radius) and the shear term only for `SOLID`. The force limit is $\Delta t \le F \min_i \sqrt{h_i/|\mathbf{a}_i|}$ with F = `FORCES_FACT`.

8.3 Fast integration scheme

With `FAST_INTEGRATION_SCHEME` (Raducan and Jutzi, 2022; Jutzi et al., 2022), the late, low-velocity stage of granular flow, where the time step is set by the high sound speed of the material although the flow is slow, can be computed with much larger time steps. At the transition time t_f of a material (group `fast_integration` in the material config), its EOS is replaced by a linear medium with a low bulk modulus K_f,

$$p = K_\mathrm{f}\left(\frac{\varrho_\mathrm{s}}{\varrho_{0,\mathrm{f}}} - 1\right) \quad \text{(Murnaghan with } n = 1\text{)}, \qquad c_\mathrm{s} = \sqrt{K_\mathrm{f}/\varrho_{0,\mathrm{f}}}, \tag{50}$$

with $\varrho_\mathrm{s} = \varrho$ for non-porous (types 1, 2, 7 $\to$ 1) and $\varrho_\mathrm{s} = \alpha\varrho$ for p-alpha materials (types 5, 6, 13 $\to$ 6, $p = K_\mathrm{f}(\alpha\varrho/\varrho_{0,\mathrm{f}} - 1)/\alpha$). The shear modulus becomes $\mu_\mathrm{f} = 3K_\mathrm{f}(1-2\nu)/[2(1+\nu)]$ with the given or the current Poisson ratio ν, and the deviatoric stress is rescaled by μ_f/μ so that the elastic strain is unchanged. The tension cut-off `rho_limit` of the new medium defaults to 1 (no tension below $\varrho_{0,\mathrm{f}}$). The strength model is unchanged. The scheme is valid only if the flow is slow compared to the new sound speed.

9 Self-gravity

Self-gravity (`-s`) is computed with a Barnes-Hut tree (Barnes and Hut, 1986) following the GPU algorithms of Burtscher (2011). The same tree is used to find the SPH interaction partners (Hernquist and Katz, 1989).

9.1 Tree construction

The hierarchical decomposition is stored in an octree (quadtree in 2D). Since pointers are inconvenient on the GPU, one index array `childList` holds all particles and tree nodes; up to $\lceil 2.5N \rceil$ nodes with 2^D children each are allocated. The root node is the cube enclosing all particles. The particles are inserted in parallel: each thread descends from the root to the leaf where its particle belongs and locks the child entry with `atomicCAS` (status -2, "locked"; -1 means "empty"). If the entry is occupied by another particle, new inner nodes with half the edge length are created until both particles reside in different leaves; then the lock is released. Afterwards, the masses and centres of mass of all inner nodes (pseudo-particles) are computed bottom-up.

9.2 Gravitational force

For each particle the tree is walked from the root. A node of edge length s at distance r is used as a whole if

$$s/r < \vartheta, \tag{51}$$

otherwise its children are opened. The acceleration from a pseudo-particle of mass m_p at $\mathbf{x}_p$ is $\mathbf{a}_i = -Gm_p\,(\mathbf{x}_i - \mathbf{x}_p)/r^3$, with a smoothed distance to avoid singularities. Only monopoles are used; therefore $\vartheta \leq 0.5$ (`-a`, default 0.5) is recommended (Barnes and Hut, 1989). Direct summation is available with `-D`.

9.3 Decoupled gravity

With `-g`, the tree gravity is not recomputed in every right-hand side evaluation. It is recomputed every tenth evaluation, or earlier if more than 0.1 % of the particles have changed their tree node since the last computation; in between, the stored gravitational accelerations are reused. This saves time if the hydrodynamic time step is much shorter than the gravitational time scale. The criterion is under revision.

9.4 Point masses

With `GRAVITATING_POINT_MASSES`, point masses interact gravitationally with each other and with the SPH particles (including the back reaction). SPH particles closer than $r_{\min}$ or farther than $r_{\max}$ are removed, or accreted with `PARTICLE_ACCRETION` and `UPDATE_SINK_VALUES`.

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